

MENGWEI HU

✉ m.hu@jhu.edu | 🌐 hmengwei.github.io

EMPLOYMENT

J.J. Sylvester Assistant Professor, Johns Hopkins University 2026–2029
Mentor: Mikhail Khovanov

EDUCATION

Ph.D. in Mathematics, Yale University 2020–2026
Advisor: Ivan Losev
Thesis: *Geometry of anti-Poisson involutions in the deformations and resolutions of Kleinian singularities*

B.Sc. (Honors) in Mathematics, Sichuan University 2016–2020

RESEARCH INTERESTS

Geometric representation theory, algebraic geometry, and symplectic geometry. In particular, modules over quantizations, involutions of quiver varieties, and categorification of zigzag algebras.

PUBLICATIONS

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- M. Hu, On certain Lagrangian subvarieties in minimal resolutions of Kleinian singularities. *Submitted*. arXiv:2504.08717.
 - M. Hu, Presentations of diagram categories. PUMP J. Undergrad. Res. **3** (2020), 1–25. DOI: 10.46787/pump.v3i0.2256.

FUNDED COLLABORATIONS

Institute for Advanced Study, Summer Collaborators Program 2026

FELLOWSHIPS & AWARDS

-
- **AMS-ICM Travel Grant**, funded by the National Science Foundation 2026
 - **AMS Sectional Travel Grant**, 2026 Spring Southeastern Sectional 2026
 - **AWM Symposium Travel Grant**, 2025 Research Symposium 2025
 - **Fan Family Fellowship**, Yale University 2020–2021

INVITED TALKS

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- AMS Fall Central Sectional (Recent Progress in Representation Theory), UMN Duluth Oct 2026
 - AMS Fall Southeastern Sectional (Geometric & Homological Representation Theory), KSU Oct 2026
 - Algebra and Number Theory Seminar, Louisiana State University Apr 2026
 - AMS Spring Southeastern Sectional (Noncommutative Algebra), Georgia Southern University Mar 2026
 - AMS Spring Southeastern Sectional (Representation Theory), Georgia Southern University Mar 2026
 - Cluster Algebras and Lie Theory Seminar, University of Notre Dame Mar 2026
 - Geometry, Physics, and Representation Theory Seminar, Northeastern University Jan 2026
 - Geometry, Algebra, and Theoretical Physics Seminar (Zoom), Imperial College London Dec 2025
 - Representation Theory Seminar, University of Massachusetts Amherst Nov 2025
 - AMS Central Sectional (Geometry, Representation Theory & Noncommutative Algebra), SLU Oct 2025

- Algebra Seminar, University of Georgia *Oct 2025*
- Topology Seminar, Johns Hopkins University *Sep 2025*
- Algebra Seminar, Syracuse University *Sep 2025*
- Algebra Seminar, University at Buffalo *Sep 2025*
- AWM Research Symposium (Mathematical Physics & Representation Theory), UW-Madison *May 2025*
- Algebra Seminar, University of Oregon *Mar 2025*
- Algebra Seminar, UCLA *Mar 2025*
- Geometry, Symmetry and Physics Seminar, Yale *Jan 2025*
- AMS Western Sectional (Geometric & Categorical Representation Theory), UC Riverside *Oct 2024*
- AMS Eastern Sectional (Singularities in Commutative Algebra), University at Albany *Oct 2024*

Contributed Talk

- International Congress of Mathematicians (ICM) 2026, Philadelphia (short communication) *Jul 2026*

RESEARCH VISITS

- University of Texas at Austin *Oct 2025*

TEACHING

Instructor of Record

- AS.110.106: Calculus I (Biology and Social Sciences), Johns Hopkins University *Fall 2026*
- MATH 115: Calculus of Functions of One Variable II, Yale University *Fall 2024, Spring 2026*
- MATH S118E: Introduction to Functions of Several Variables, Yale University *Summer 2024*
- MATH 112: Calculus of Functions of One Variable I, Yale University *Fall 2022, Fall 2023*

Certificate of College Mathematics Teaching Preparation, Yale University *2025*

Teaching Assistant: Linear Algebra, Calculus, Yale University *Spring, Summer, Fall 2021*

MENTORING

- **REU Mentor**, Summer Undergraduate Math Research at Yale *Summer 2025*
“Superalgebras and Polynomial Spaces” (with Dwight Williams II)
- Mentor, Association for Women in Mathematics Mentor Network *2025–present*
- Mentor, Yale Math Graduate Buddies Program *Fall 2022*
- Mentor, Yale Take an Undergraduate to School *Spring 2022*

SERVICE AND OUTREACH

Journal Referee

- *Representation Theory* *2025–present*

Seminar Organization

- Co-organizer, Johns Hopkins Topology Seminar *Fall 2026*
- Co-organizer, Joint Yale-Columbia Representation Theory Seminar *Fall 2022*
- Co-organizer, Yale Graduate Student Seminar *Spring 2022*

Panels, Committees, and Outreach

- Panelist, Women in Mathematics in New England (WIMIN), Graduate School Panel *Sep 2024*
- Panelist, Lang Teaching Seminar at Yale, Graduate Instructor Panel *Mar 2024*
- Committee Member, Yale Graduate Student Advisory Committee *2023–2026*